Ecological Indicators 141 (2022) 109110

Contents lists available at ScienceDirect

Ecological Indicators

journal homepage: www.elsevier.com/locate/ecolind

Predicting land change trends and water consumption in typical arid regions using multi-models and multiple perspectives

Qingling Bao [a,b], Jianli Ding [a,b,*], Lijing Han [a,b], Jiang Li [c], Xiangyu Ge [a,b]

[a] College of Geography and Remote Sensing Sciences, Xinjiang University, Urumqi 800046, China
[b] Key Laboratory of Smart City and Environment Modelling of Higher Education Institute, Xinjiang University, Urumqi 800046, China
[c] Department of Water Resources of Xinjiang Uygur Autonomous Region, Urumqi 800046, China

ARTICLE INFO

Keywords:
Patch-generating land use simulation model
Google Earth Engine
Land Cover
Arid Region
Saline Land
Water Consumption

ABSTRACT

Water scarcity has emerged as a major impediment to the long-term development of inland arid region basins. Increased human activity has produced land degradation in arid areas and water stress in recent years. The objective of this research was to expose patterns in land cover changes in inland arid basin and the mechanisms driving them, as well as to optimize the socioeconomic and ecological water consumption structure of water resources. Therefore, this research used Google Earth Engine to map land cover year by year from 2000 to 2020, simulated future land cover changes and driving force analysis using a Patch-generating land use simulation model, and finally established the relationship between water consumption of different systems and different land cover types using linear equations, and scenario simulation method were used to estimates the ecological and agricultural water consumption in 2035. The results showed (1) Over the previous 20 years, there has been an increase in the amount of saline, agricultural land, and natural vegetation (grassland, scrub, and forest), as well as a considerable drop around permanent glaciers and snow. (2) Under the Markov-2035 scenario, saline land decreases to 32.92 km^2, natural vegetation decreases to 381.89 km^2, arable land shrinks to 135.69 km^2, and the permanent glacier shrinks to 30.03 km^2. Ecological water consumption would be 0.127×10^8 m^3, while agricultural water consumption would be 0.394×10^8 m^3. (3) Elevation influenced the variety of all land cover types. Saline land was more sensitive to temperature, railroad proximity, and pH; agricultural land was more sensitive to population density and gross regional product; and natural vegetation was more sensitive to soil organic matter, railroad proximity, and temperature.

1. Introduction

For several years, sustainable development in inland arid regions has been an international priority (Chinnasamy et al., 2021; Wang et al., 2010). A number of environmental issues such as soil degradation, soil erosion, excessive deforestation, water pollution, salt and duty tempests, lake shrines, soil salinization, and desertification have led to an increase in the ecology and socio-economic water pressure due to high intensity human activity (Liu et al., 2010; Pokhrel et al., 2021). The emergent effect on sustainable development has clearly been negative (Tumbare, 2021). The most direct manifestation of the above environmental problems is the dynamic change of land cover, which is the carrier of the above environmental issues and an important part of global environmental change (Chen et al., 2019). Anthropogenic activities and climate change can both motivate and limit changes in land cover (Santos et al., 2019). Rapid economic development has caused rapid growth of agricultural land and cities at the expense of forests, grasslands and wetlands, which in turn have disappeared causing even worse ecological conditions and making the ecological environment of the existing fragile arid regions even more unsuitable for human livelihoods (Cooley et al., 2021; Feng et al., 2016; Li et al., 2018). Reasonable allocation and real-time scheduling of water resources are extremely important for arid region basin management (Huang et al., 2014). Since the reform and development, China has entered the stage of rapid economic development, and the allocation of water resources was dominated by the economic development planning department, ignoring the ecological water demand (Bryan et al., 2018; Lu et al., 2018; Wang et al., 2020). In addition, the large-scale abuse and unscientific exploitation of water resources have worsened the ecological environment of the already fragile interior arid regions (Zhang et al., 2021a).

* Corresponding author at: College of Geography and Remote sensing Sciences, Xinjiang University, Urumqi 800046, China.
E-mail addresses: baoqingling@stu.xju.edu.cn (Q. Bao), watarid@xju.edu.cn (J. Ding).

https://doi.org/10.1016/j.ecolind.2022.109110
Received 2 February 2022; Received in revised form 23 May 2022; Accepted 24 June 2022
Available online 28 June 2022
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Because traditional field investigation is inefficient and expensive, many researchers have recently used high-resolution remote sensing techniques, which can be accessed in real time, to uncover and analyze landcover patterns (Gong, 2019; Liao et al., 2017; Nasta et al., 2017). Remote sensing satellite data are a particularly useful source of information for the land cover classification, the resolution of restrictions on hard-to-reach field areas and the provision of real-time mapping. Data from Landsat series and Sentinel series are frequently utilized for land

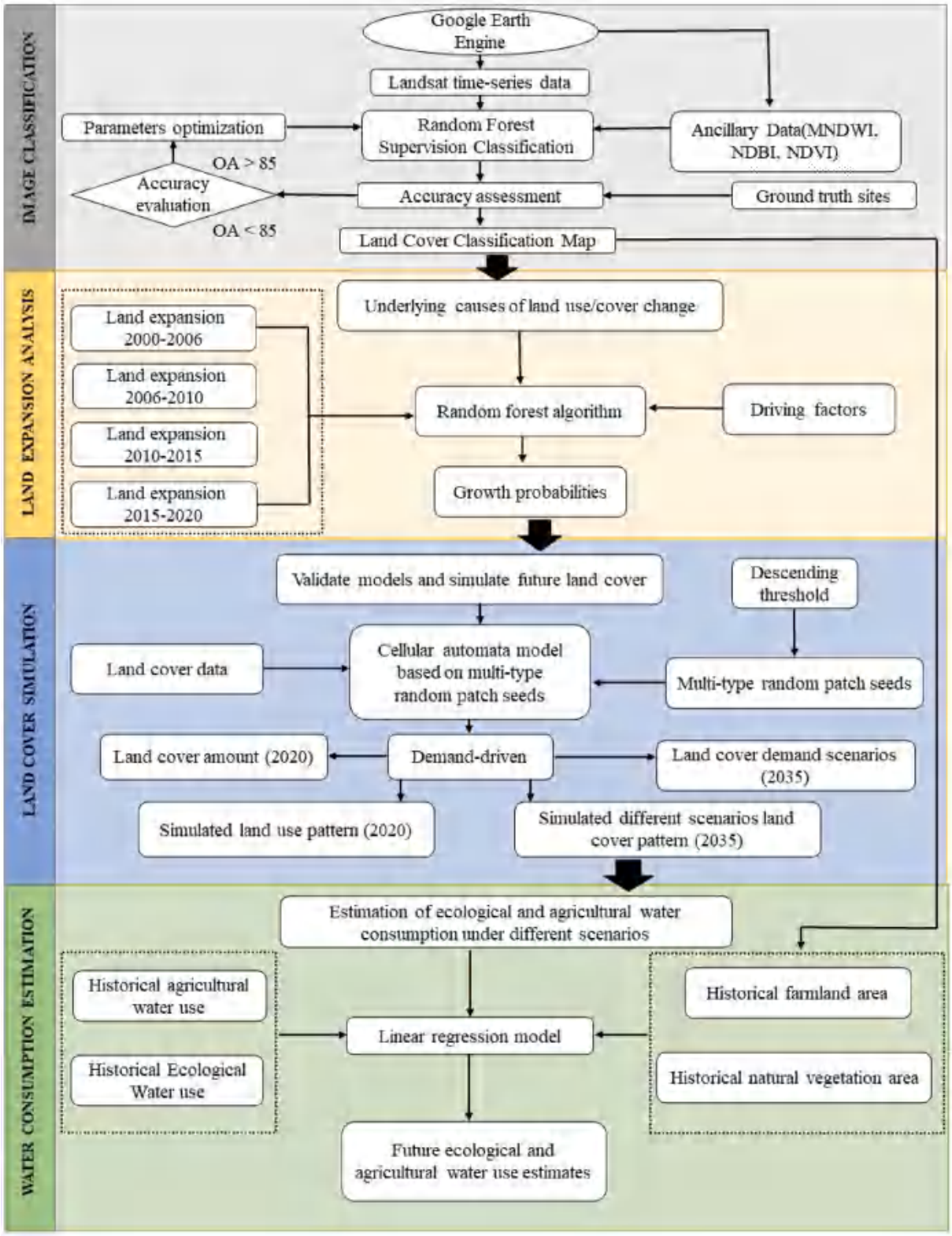

Fig. 1. Workflow diagram of this study. The first part shows the GEE classification process to obtain time series land cover data. The second part shows the drivers behind land cover change. The third section shows the simulation and validation of future land cover using the PLUS model to obtain the spatial and temporal distribution and area of land cover in 2035. The fourth section shows the use of linear models to estimate future ecological and agricultural water consumption.

cover analyses (Lin et al., 2021). Inside the Ebinur Lake Basin around 2018, Wang et al. (2019) used Landsat and Sentinel-1A data to analyze seasonal dynamics. Yushanjiang et al. (2018) used Landsat data to investigate the relationship between land cover and ecosystem service values in the Lake Ebinur basin. Ge et al. (2019) used Landsat data to map saline lands in the Lake Ebinur basin. Zhu et al. (2021) studied the relationship between carbon storage and land cover change and found that natural vegetation degradation was the main cause of the decrease in carbon storage. Yang et al. (2021) assessed the impact of land cover changes on Net Primary Productivity and its drivers in China. The research presented above provide a cursory examination of the scientific subject of land cover dynamics in inland arid basins, however, concerns such as lengthy time series, large scale, driving forces, and future changes in land cover remain unresolved (An et al., 2017). With the rapid development of big data and cloud computing technologies in recent years, an increasing number of researchers have used Google Earth Engine to map land cover changes, saving data processing time and improving efficiency, and enabling dynamic real-time mapping that was previously impossible (Phan et al., 2020; Sidhu et al., 2018; Zhang et al., 2019b). Clarifying ecological and economic water consumption and optimizing water resource regulatory mechanisms are linked to regional people's well-being for the sustainable development of arid regions (Rost et al., 2008). Yuan and Shen (2013) used the Penman-Monteith method and meteorological data to estimate agricultural water consumption. Su et al. (2021) found that high water consumption in agriculture leads to the continuous shrinkage of the Aral Sea through land cover datasets and related statistics. Wu et al. (2021) found that agricultural irrigation water consumption increased with the degree of drought through a modified Soil and Water Assessment Tool. Water stress research in arid regions has primarily relied on technical techniques such as hydrological models, statistical approaches, and remote sensing, none of which can provide predictions of water consumption for future scenarios.

Previous research mostly tracked the dynamic of land cover without linking it to water consumption. The effects on land cover of human activities and natural forces remain uncertain. The objective of this work ought to be to identify factors that drive change in farmland and other types of land cover in inland arid areas, then optimize the structure for the allocation of water resources. The present study employs the Google Earth engine and a patch generation land use simulation model to track long-term land cover changes in a typical inland arid region in the Lake Ebinur basin and to forecast future land cover trends. Estimation of future agricultural and ecological water consumption by the development of linear equations based on simulation findings for various land cover scenarios. Differentiate quantitatively the influence of natural and human forces on land cover change.

2. Materials and methods

This study attempts to use the Google Earth Engine (GEE) platform to monitor the dynamics of the long time series of land cover in the typical inland arid region of Lake Ebinur basin, use the Patch-generating land use simulation model (PLUS) to simulate future patterns, explore the driving forces of their changes, to finally estimate the current and future scenarios of agricultural and ecological water consumption (Fig. 1).

2.1. Study area

The Ebinur Lake basin is located in the southwest section of the Junggar Basin in Xinjiang, China, approximately 60 km north to south and 110 km west to east, with a total area of around 53,584 square kilometers (Jing et al., 2020). Because the region has a continental desert climate, which results in limited precipitation throughout the year, extreme temperature variations, and up to an average annual evapotranspiration of more than 2000 mm, the arid climate conditions contribute to the region's scarcity of water resources (Wang et al., 2017). The region's main rivers are the Jing and Bortala, which pour into the desert depression from the mountain outlet and eventually form the coccyx, Lake Ebinur (Ma et al., 2007). The basin is characterized by desert flora including camel thorn, pike, and red willow, as well as artificial vegetation like cotton, wheat, and corn (Fig. 2). The basin plain area's typical regional soils include gray-brown desert soil and gray-desert soil, with wind-sand soil being the most extensively distributed. In some locations, salinization and desertification are more severe (Zhang et al., 2019a). The Lake Ebinur basin's poor natural condition is a result of limited water resources. The surface area of Lake Ebinur has dropped dramatically since the 1950s, yet the population has expanded significantly, resulting in a slew of environmental and ecological issues such as saline land and desertification. The shrinkage of Lake Ebinur has exposed an extensive bare lake bed and has been reconsidered as one of the sources of sand and salt dust in Central Asia (Zhang et al., 2015).

2.2. Data sources

In this study, three key categories of data were used: satellite images, socioeconomic drivers, and natural environment drivers (see Table 1). Satellite images with a time range of 2000 to 2020 and a spatial resolution of 30 m × 30 m were retrieved from Google Earth engine as the data source for land cover classification. Table 1 shows the socioeconomic and natural environment data that are used to investigate the causes of changes in land cover.

2.3. Image classification

2.3.1. GEE image collection

Landsat series satellites with extended time series were chosen as data providers due to their long-time range. To build the image collection, all Landsat 5 TM, Landsat 7 TM, and Landsat 8 OLI images covering the whole Ebinur Lake basin were filtered in GEE's code editor (Java) interface. Time, cloud mask, and position filtering are the conditions for filtering the data. The time circumstances were chosen based on the quality of satellite pictures taken between 2000 and 2020, from April to August of each year. The classification used an average of 80 Landsat pictures per year. An algorithm offered by GEE was used to de-cloud the cloud masks for Landsat data. The band properties utilized for classification were chosen for each sensor's visible, near infrared, and short-wave infrared bands. To better discern agricultural fields, the Normalized Difference Vegetation Index (NDVI) was generated for all images in the image collection (Table 2) (Carlson and Ripley, 1997). Normalized Difference Built-up Index (NDBI) and Modified Normalized Difference Water Index (MNDWI) are calculated to classify built-up land and water bodies (Table 2) (Singh et al., 2015; Zhang et al., 2009).

2.3.2. Classification model

GEE includes a number of machine learning supervised classification techniques, such as Random Forest, Bayesian, and Support Vector Machine (Shelestov et al., 2017). The classifier in this work was the widely used random forest machine learning technique, which was used to rectify the habit of overfitting the decision tree to its training set. The number of trees in the study's model was set at 20. In 2011, 2015, and 2020, polygon feature datasets were constructed from local land cover data and visual interpretation, encompassing water bodies, salt-affected areas, Agriculture, Barren, Grassland/Shrubland, Forest, Ice/Snow, and urban building lands, respectively (Table 3).

2.4. The patch-generating land use simulation model (PLUS)

The patch-generating land use simulation model (PLUS) multiclass land use of simulation model, which was developed by Liang i.e. in 2021 (Liang et al., 2021). The model is divided into four major modules: (1) a rule mining module based on the land expansion analysis strategy (LEAS); (2) a module based on multi-type stochastic patch seed *meta-*

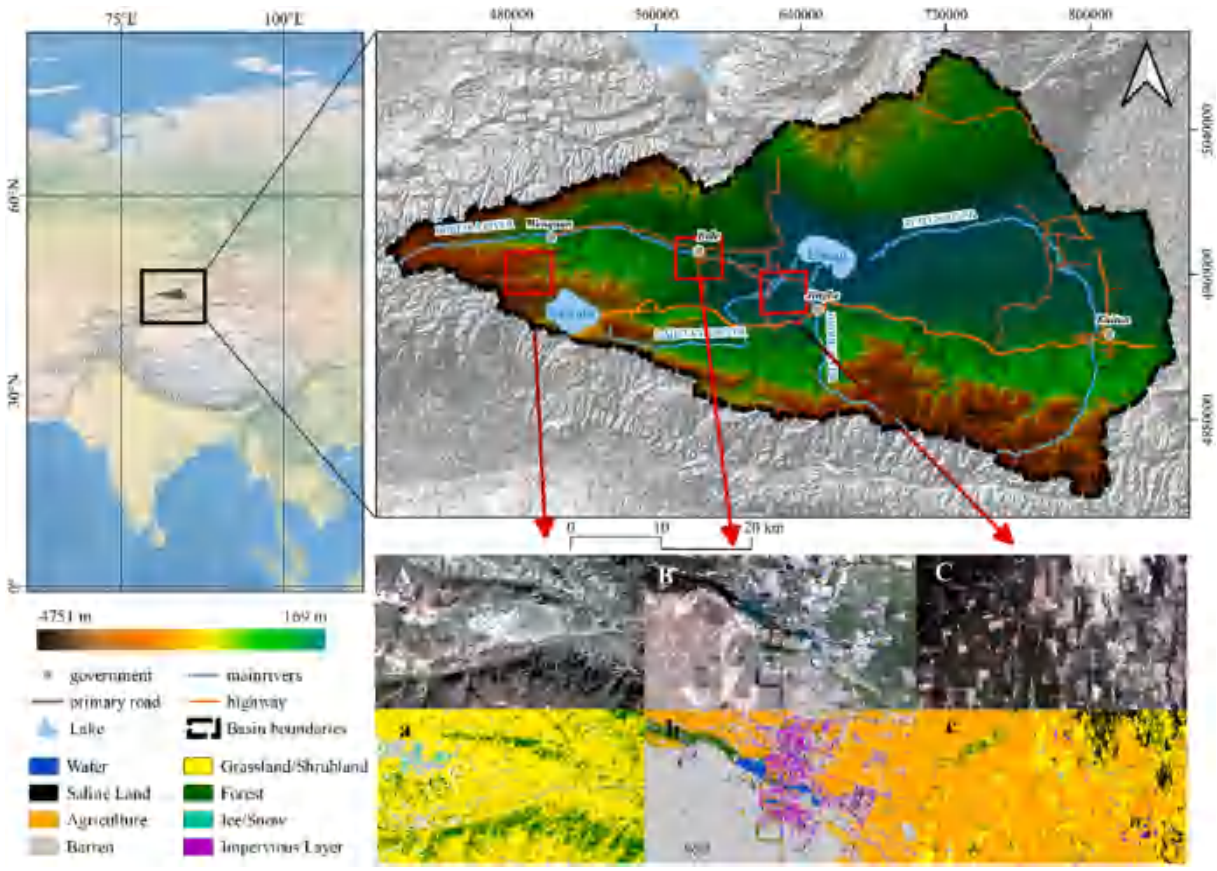

Fig. 2. Geographical location of the study area (A, B, and C represent the true color composite of Landsat images, respectively. a, b, and c represent the Google Earth engine classification results for the high-elevation, urban, and agricultural desert interlacing zones in the study area, respectively.).

Table 1
Remote sensing and spatial driving data used in this study.

Type	Data	Year	Resolution(m)	Data source
Remote Sensing Data	Landsat 7 Surface Reflectance	2000–2002	30	https://developers.google.com/earth-engine/datasets/catalog/
	Landsat 5 Surface Reflectance	2006–2011	30	https://developers.google.com/earth-engine/datasets/catalog/
	Landsat 8 Surface Reflectance	2013–2020	30	https://developers.google.com/earth-engine/datasets/catalog/
Socio-economically driver	Proximity to government	2020	100	https://map.baidu.com/
	Proximity to highway	2020	100	https://www.openstreetmap.org/
	Proximity to railway	2020	100	https://www.openstreetmap.org/
	Proximity to primary road	2020	100	https://www.openstreetmap.org/
	Proximity to secondary road	2020	100	https://www.openstreetmap.org/
	Proximity to tertiary road	2020	100	https://www.openstreetmap.org/
	Proximity to trunk road	2020	100	https://www.openstreetmap.org/
	Proximity to highway	2020	100	https://www.openstreetmap.org/
	GDP	2010	1000	http://www.geodoi.ac.cn/WebCn/Default.aspx
	Population	2010	1000	http://www.geodoi.ac.cn/WebCn/Default.aspx
Nature and Environment Driver	Proximity to rivers	2015	100	https://www.hydrosheds.org/downloads
	Precipitation (mm)	2000	1000	https://www.worldclim.org/data/worldclim21.html
	Solar radiation (kJ m^{-2} day^{-1})	2000	1000	https://www.worldclim.org/data/worldclim21.html
	Average temperature (°C)	2000	1000	https://www.worldclim.org/data/worldclim21.html
	Soil PH	2000	1000	Shangguan et al.,2013
	Soil Bulk Density	2000	1000	Shangguan et al.,2013
	Soil Clay Content	2000	1000	Shangguan et al.,2013
	Soil Gravel Content	2000	1000	Shangguan et al.,2013
	Soil Sand Content	2000	1000	Shangguan et al.,2013
	Soil Silt Content	2000	1000	Shangguan et al.,2013
	Soil Porosity	2000	1000	Shangguan et al.,2013
	Soil Organic Matter	2000	1000	Shangguan et al.,2013
	Soil Total K	2000	1000	Shangguan et al.,2013
	Soil Total N	2000	1000	Shangguan et al.,2013
	Soil Total P	2000	1000	Shangguan et al.,2013
	DEM		90	https://www.hydrosheds.org/downloads
	Slop		90	

Table 2
Information of remote sensing index.

Indicators	Equation	Features
NDVI	(NIR−R)/(NIR + R)	It has been applied to detect vegetation growth status, vegetation cover and eliminate some radiation errors.
NDBI	(MIR−NIR)/(MIR + NIR)	Extraction of building and impermeable layer information
MNDWI	(Green−MIR)/(Green + MIR)	Extraction of water body information

Note: NIR, R, MIR and Green represent the near-infrared band, red band, mid-infrared band, and green band respectively.

Table 3
Land Cover Classification System.

Land cover class	Description
Water	Permanent open water bodies, such as lakes, reservoirs, rivers, etc.
Saline Land	Land eroded by soil salinity
Agriculture	Farmland and pasture, generally growing cotton and wheat
Barren	exposed earth surface, no vegetation growing
Grassland/ Shrubland	Land where vegetation type is mainly grasses and scrub
Forest	A forest is an area of land dominated by trees.
Ice/Snow	A land covered with snow and glaciers all year round
Impervious Layer	Villages and Towns

cellular automata (CARS); and (3) a module for predicting future land use demand (including linear and Markov methods). The land class expansion maps of land cover data from 2000 to 2006, 2006–2010, 2010–2015, and 2015–2020 were extracted first in the LEAS module, followed by an analysis of the most influential drivers of each land class expansion using the random forest algorithm, and finally the growth probability maps of each land class in each period were derived. Land cover was simulated from the macro-scale (target land cover amount) to the micro-scale (each grid land cover type) in the CARS module, using the growth probability maps derived from the LEAS module as input, and based on pre-defined future land cover linear projections and Markov chain projection scenarios, as well as embedded competition mechanisms. The PLUS model's Markov chain and linear approaches can be used to estimate the quantity of future land cover. In this study, the spatial distribution of land cover and the area of each land class in 2035 were estimated using the methods described above.

2.5. Accuracy Assessment

The following parameters were used for accuracy validation during both GEE classification and PLUS model simulations: overall accuracy (OA), kappa coefficient (KC) and user accuracy (UA). OA is the total classification percentage, which is the ratio of the number of correctly classified units to their total number; KC represents the proportion of evaluated classifications that produce fewer errors than completely random classifications; and UA represents the ratio between correct classifications and all classified units. The above parameters range from 0 to 1, with larger values representing better classification and simulation results.

$$OA = \sum_{i=1}^{n} \frac{P_{ii}}{P} \tag{1}$$

$$UA = \frac{P_{jj}}{P_{+j}} \tag{2}$$

$$KC = \frac{OA - \frac{\sum_{i=1}^{n}(P_{i+}P_{+i})}{\sum_{i=1,j=1}^{n} P_{ij}}}{1 - \frac{\sum_{i=1}^{n}(P_{i+}P_{+i})}{\sum_{i=1,j=1}^{n} P_{ij}}} \tag{3}$$

where P_{ij} and P represent the total number of correctly classified pixels and the total number of all pixels involved in the classification, respectively. P_{jj} and P_{+j} represents the number of correctly classified pixels in a certain land cover/use type and the predicted amount of all pixels of this type, respectively. n, i and j represent the number of all land cover/use types, measured types, and predicted types, respectively. Pi+ represents the sum of data of actual type.

3. Results

3.1. Model performance

We examined the OA, KC, and UC of the GEE land cover categorization results year by year from 2000 to 2020 using field validation sample sites, while remote sensing data from 2001 to 2005 and 2012 were missing. To validate the PLUS model's effectiveness for land cover simulation, we used land cover data from two periods, 2015 and 2020, extracted maps of the probability of growth of each class, simulated the land cover in 2020 with 2015 as the baseline and 2020 as the target for each class demand, and validated with the actual 2020 land cover. The results reveal that the overall classification effect of the GEE random forest method performed well, with OA above 92 percent, KC remaining stable at around 92 percent, and UA consistently over 85 percent. Meanwhile, the overall correctness of the PLUS model simulation has reached the level of confidence. The lowest OA was in 2007_L5 and the greatest was in 2011 L5; the lowest KC was in 2016_L8 and the highest was in 2007_L5; and the lowest UA was in 2015_L8 and the highest was in 2013. For OA, KC, and UA, the PLUS model was simulated with 87.05 percent, 81.51 percent, and 82.85 percent, respectively (Table 4).

The FROM-GLC10 product classified by Gong et al. (2019) was taken for validation, after reclassification and resampling to the LULC data format of this study, as the saline land classified in this paper does not exist in the product, so the classification results of this paper were reclassified for comparison convenience. Fig. 3 shows the spatial and quantitative comparison of the two land cover data in 2017. We can find that the results based on GEE classification have a large difference with the FROM-GLC10 product in impervious layer classification, but in other

Table 4
A summary of Landsat classification accuracy from 2000 to 2020.

year	Overall accuracy (%)	Kappa statistic (%)	User's accuracy (%)
2000_LAN7	95.96	93.74	86.34
2001_LAN7	95.03	92.23	87.44
2002_LAN7	95.28	92.64	89.98
2006_LAN7	95.44	92.93	90.45
2007_LAN5	92.36	95.21	84.12
2008_LAN5	95.95	93.73	81.87
2009_LAN5	95.5	93.02	83.45
2010_LAN5	95.52	90.95	86.25
2011_LAN5	96.68	93.18	87.1
2013_LAN8	94.94	92.65	90.58
2014_LAN8	94.62	92.24	85.88
2015_LAN8	95.16	93.82	75.81
2016_LAN8	93.29	90.47	79.98
2017_LAN8	95.3	92.96	83.96
2018_LAN8	94.95	92.69	86.75
2019_LAN8	94.82	92.49	78.85
2020_LAN8	94.92	92.65	89.96
2020_SIM	87.05	81.51	82.85

Note: LAN5, LAN7 and LAN8 are abbreviations of landsat-5, landsat-7 and landsat-8 respectively, which represent the remote sensing images used in that year. All remote sensing data are from March to August of that year.

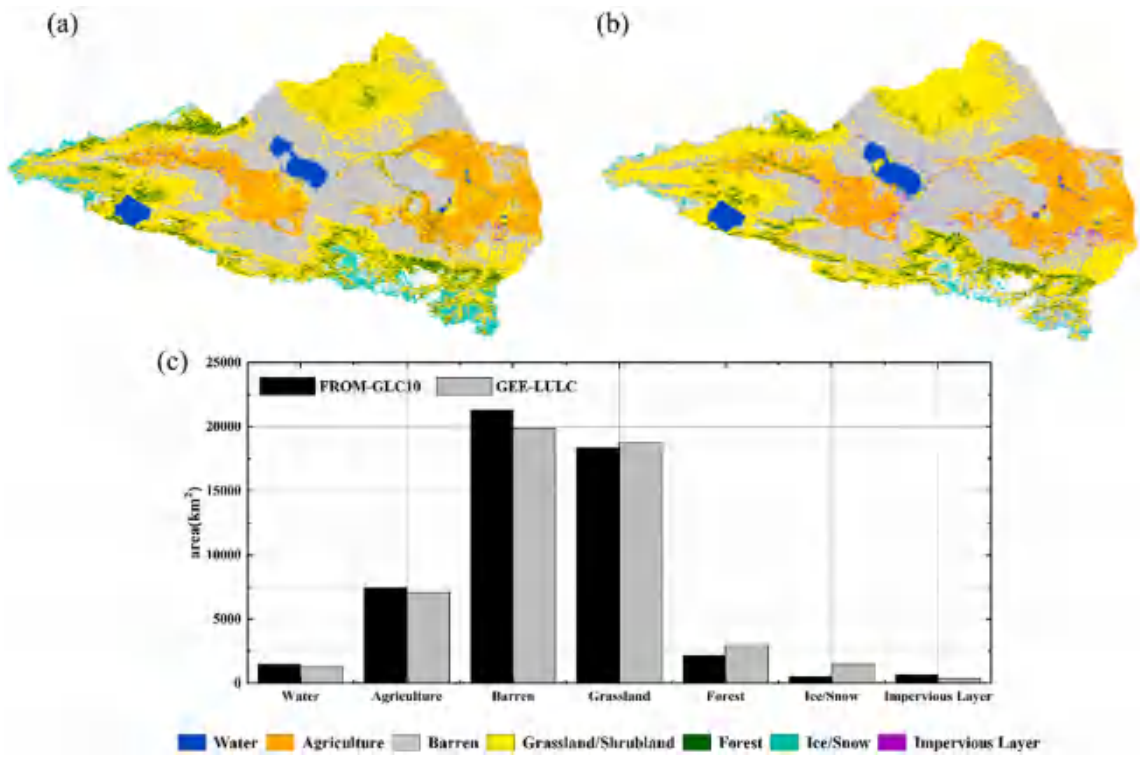

Fig. 3. Comparison of GEE classification results with existing products. Where Figure (a) shows the results based on GEE classification, Figure (b) shows the FROM-GLC10 land cover product, and Figure (c) shows the quantitative comparison of both land cover data.

land classes are basically consistent with the FROM-GLC10 product in spatial and quantitative terms. The overall Kappa Accuracy, Overall Accuracy, Producer's Accuracy, and User's Accuracy were found to be higher than 0.65, which is a good level when compared with the FROM-GLC10 product (Table 5). However, Producer's Accuracy and User's Accuracy are lower in forest and impervious layer and need to be improved.

3.2. The land cover/use changes from the GEE classification results

We obtained long time series land cover maps of the study area and statistics on the area share of each class based on the GEE classification results (Fig. 4 & Fig. 5). The results showed that the area of saline land, water bodies, and forests fluctuated, whereas the area of agricultural land, grassland/scrubland increased from year to year, the area of bare land declined marginally, and the area of glaciers decreased significantly. We discovered that saline land surrounds the tailrace lake and extends to the southwest; agricultural land was primarily concentrated in the floodplain of the basin and developed around the river and the lake; forests, grasslands, and scrubland accumulated in the basin's mountainous areas and were interwoven with agricultural lands; and glaciers and permanent snow cover were primarily located in the basin's mountainous areas.

Table 5
Classification results and product accuracy verification.

Type	Kappa Accuracy	Overall Accuracy	Producer's Accuracy	User's Accuracy
Water	–	–	0.88	0.97
Agriculture	–	–	0.80	0.84
Barren	–	–	0.79	0.85
Grassland/ Shrubland	–	–	0.72	0.71
Forest	–	–	0.62	0.56
Ice/Snow	–	–	0.88	0.69
Impervious Layer	–	–	0.19	0.17
Overall	0.76	0.85	0.69	0.68

The saline area grew from 1686.25 km^2 in 2000 to 1893.70 km^2 in 2020, with a growth rate of 10.35 $km^2 \cdot year^{-1}$. The largest increase was observed from 2006 to 2008, with an increase of about 710.89 km^2, and a cliff decline from 2016 to 2017, with a decrease of about 786.62 km^2. The area of saline increase was concentrated around the coccyx lakes of the basin and expanded to the southwest and agricultural land. The area of agricultural land increased from 4462.89 km^2 in 2000 to 6687.83 km^2 in 2020, with a growth rate of 111.24 $km^2 \cdot year^{-1}$. The growth was slow from 2000 to 2008, with a rate of about 76.84 $km^2 \cdot year^{-1}$; it increased rapidly from 2009 to 2014, with a rate of about 274.17 $km^2 \cdot year^{-1}$; and showed a fluctuating increase after 2014. Agricultural land was concentrated on both sides of the plain rivers, along the Jing, Bortala and Kuitun rivers, and gradually encroached on bare land and grassland. The total area of natural vegetation (grassland, scrub and forest) increased from 20065.48 km^2 in 2000 to 23287.33 km^2 in 2020, with an average growth rate of 161.09 $km^2 \cdot year^{-1}$. Among them, the growth was relatively slow from 2000 to 2008, with a rate of about 164.59 $km^2 \cdot year^{-1}$; from 2008 to 2009, it showed a stepwise decline, decreasing by about 2378.89 km^2; from 2011 to 2014, it showed a substantial growth trend, with a growth rate of about 1490.92 $km^2 \cdot year^{-1}$; after 2014, it showed a fluctuating trend of change. Natural vegetation was mainly concentrated in the middle and high elevation areas of the basin, and gradually expanded to lower elevation areas and agricultural lands. The area of glaciers and permanent snowpack decreased from 2681.40 km^2 in 2000 to 1,446.30 km^2 in 2020, with an average rate of decline of 61.75 $km^2 \cdot year^{-1}$. Among them, 2000 to 2011 showed an upward trend when fluctuating, with a growth rate of 62.48 $km^2 \cdot year^{-1}$; from 2013 to 2011, it showed a precipitous downward trend, with an area decrease of 1814.65 km^2; after 2013, it showed a fluctuating downward trend.

3.3. Estimation of land cover under different scenarios

We evaluated the demand for different land cover types in 2035

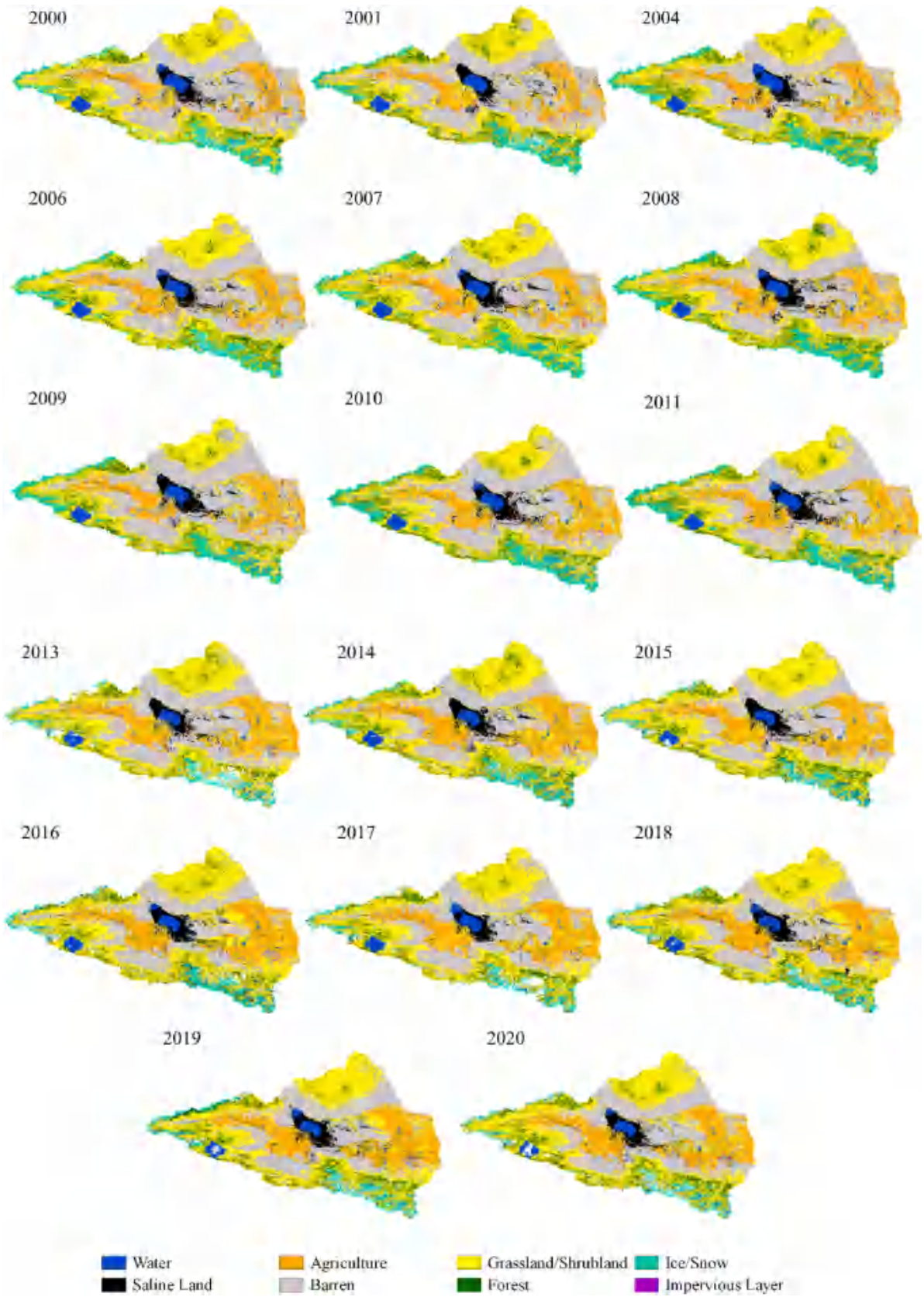

Fig. 4. Results of GEE random forest algorithm land cover classification in the Ebinur Lake basin.

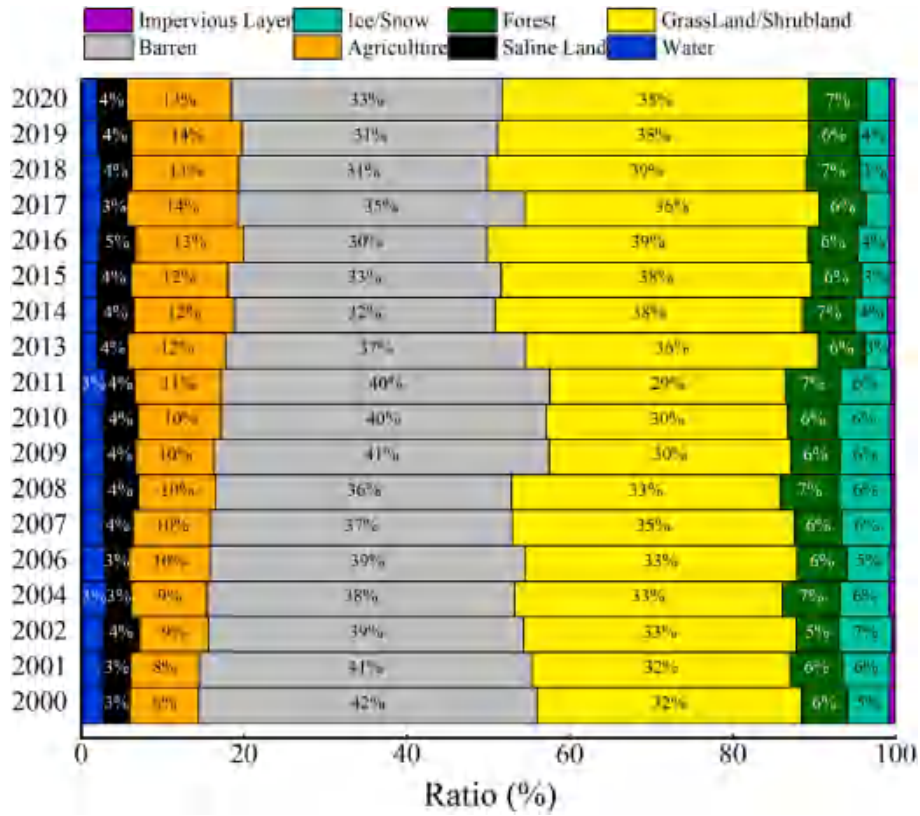

Fig. 5. Proportion of area by land cover type from 2000 to 2020.

using linear models and Markov chain approaches based on the PLUS model, and we simulated the spatial land cover data in 2035 using the CARS module. Table 6 showed the area of each land cover type under the baseline scenario, the area in 2035 using the linear model, and the area in 2035 using the Markov chain method. Fig. 6 shows the map of spatial expansion of each land cover type in each different period. Fig. 7 shows the spatial simulation maps of land cover in 2035 under two different methods estimation.

We found that water bodies decreased by 4.70 km^2 in the Linear-2035 scenario compared to the baseline scenario, while the Markov-2035 scenario increased by 3.13 km^2. Saline land increased by 1.38 km^2 in the Linear-2035 scenario, while the Markov-2035 scenario decreased by 3.05 km^2 toward the east of the basin for expansion. Agricultural land increases by 38.53 km^2 in the Linear-2035 scenario and by 17.47 km^2 in the Markov-2035 scenario, with the expansion areas concentrated on the southwest side of Lake Ebinur, both sides of the Bortala River, and both sides of the Kuitun River to the east. Bare land decreases by 66.91 km^2 in the Linear-2035 scenario and by 17.45 km^2 in the Markov-2035 scenario. Grassland and scrubland increase by 49.54 km^2 in the Linear-2035 scenario and decreases by 2.53 km^2 in the Markov-2035 scenario, with the expansion areas concentrated in the north and west of the basin. The expansion area was mainly concentrated in the north and west of the watershed. Forests decrease by 0.31 km^2 in the Linear-2035 scenario and by 1.85 km^2 in the Markov-2035 scenario, with expansion concentrated in the mountainous areas to the south of the basin. The expansion of grassland and scrubland in the Markov-2035 scenario was concentrated in the northern plateau of the basin, the western mountainous region and within the farmland; the expansion of grassland and scrubland in the Linear-2035 scenario increased, and the expansion area was more evenly distributed spatially. Glaciers and permanent snow cover decreased by 30.33 km^2 in the Linear-2035 scenario and by 2.61 km^2 in the Markov-2035 scenario. Impervious cover increased by 6.91 km^2 in the Linear-2035 scenario and decreased by 0.18 km^2 in the Markov-2035 scenario. The expansion area of glaciers and permanent snow cover in 2000–2020 was concentrated in the southern and western mountains, and the expansion area was larger in Markov-2035 scenario compared to Linear-2035 scenario. Through Fig. 7, we clearly found that the saline expansion area was much larger in the Linear-2035 scenario than in the Markov-2035 scenario and the baseline scenario. Agricultural land and natural vegetation expand the most in the Linear-2035 scenario, and glaciers and permanent snow almost disappear in the Linear-2035 scenario.

Table 6
Land cover linear prediction Markov prediction results (unit:km^2).

Type	Baseline −2019	Linear-2035	Markov-2035
Water	18.44	13.74	21.57
Saline Land	35.97	37.35	32.92
Agriculture	118.22	156.75	135.69
Barren	273.62	206.71	260.17
Grassland/Shrubland	333.26	382.80	330.73
Forest	53.01	52.70	51.16
Ice/Snow	32.64	2.31	30.03
Impervious Layer	6.91	9.22	6.73

3.4. Estimation of ecological and agricultural water consumption

To investigate the relationship between agricultural water consumption and agricultural land area, as well as ecological water consumption and natural vegetation, we obtained agricultural and ecological water consumption data from Bortala Mongol Autonomous Prefecture via the Xinjiang Uygur Autonomous Region's water resources bulletin, as well as GEE land cover classification time series data (Table 7 & Fig. 8).

In Fig. 8(a), we find that agricultural water consumption and average agricultural water consumption per hectare showed a cliff-like rise until 2010 and reached a peak in 2008, followed by a rapid decline to a minimum in 2010 and then a gradual increase, but total agricultural

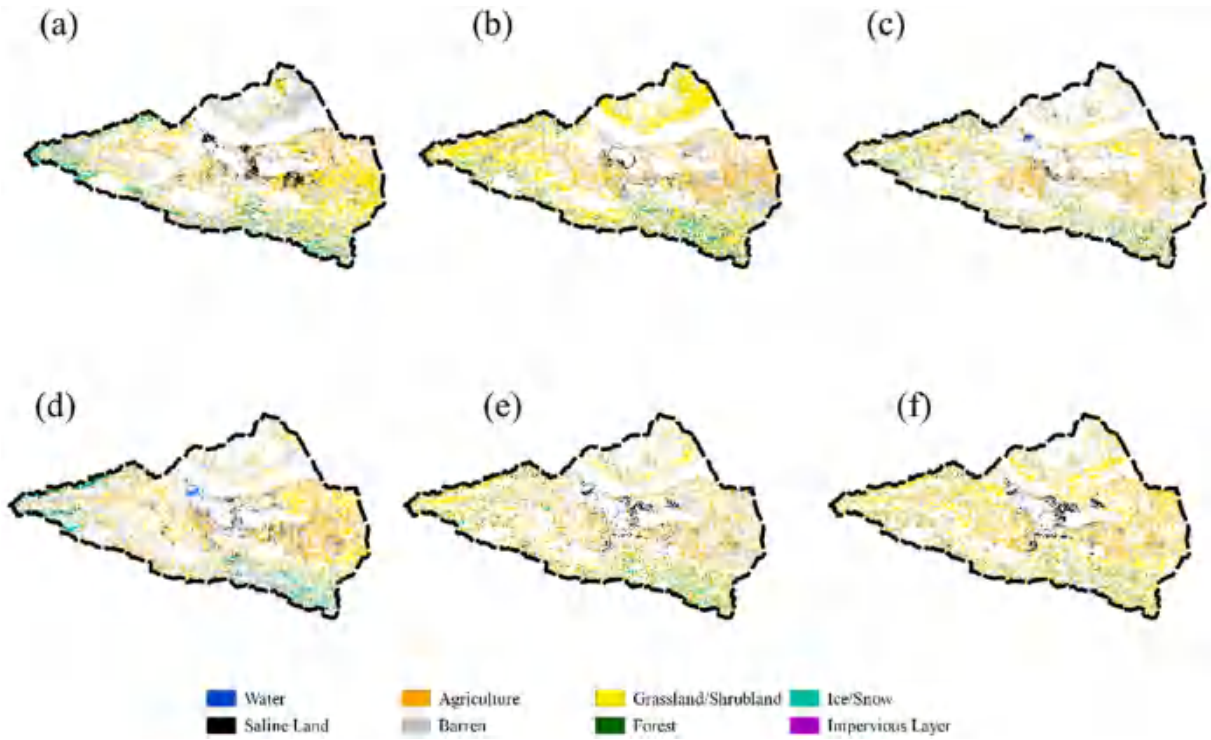

Fig. 6. Land cover expansion maps for different periods. (a) 2000–2006; (b) 2006–2010; (c) 2010–2015; (d) 2015–2020; (e) Markov Chain 2020–2035; (f) Linear estimation 2020–2035.

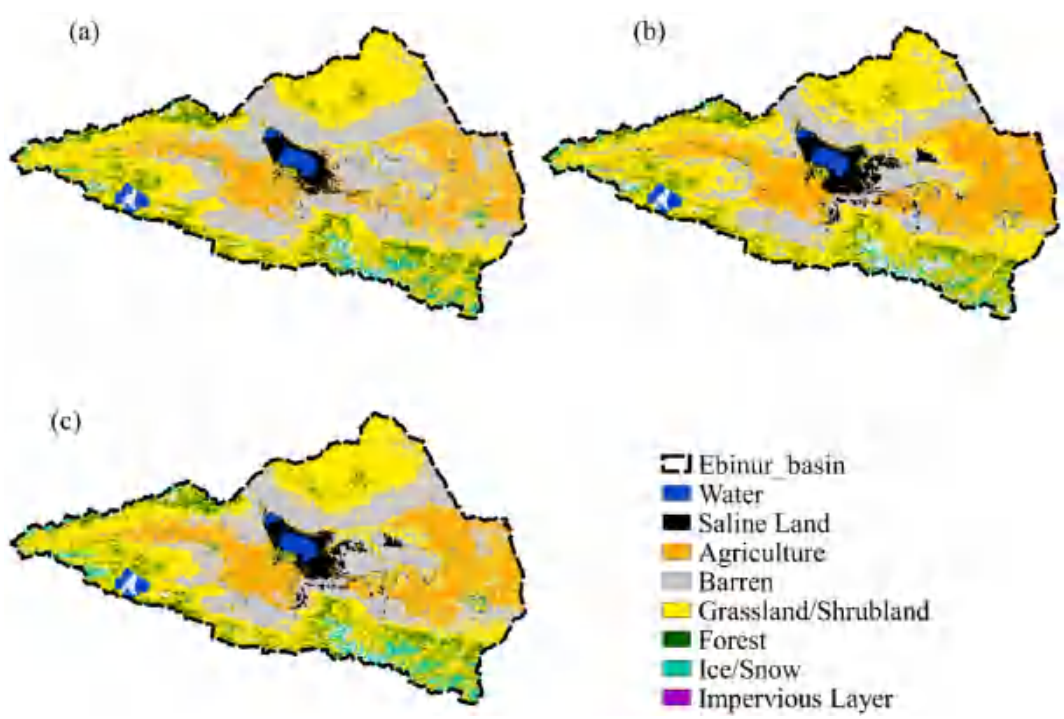

Fig. 7. Simulation of Land Cover Future Scenario Outcomes. (a) Baseline 2020 land cover; (b) Linear trend estimation of land cover in 2035; (c) Markov chain estimates of land cover in 2035.

water consumption increased faster than average agricultural water consumption per hectare. The farmland area continued to increase and was consistent with the trend of agricultural water consumption after 2010. In Fig. 8(b), the linear equation of farmland area and agricultural water consumption was modeled, and the correlation coefficient and R^2 both reached above 0.80, which proved the model credible. In Fig. 8(c, d), we found that ecological water consumption and natural vegetation area showed an increasing trend in the same way, and the fluctuation of farmland area was larger than that of ecological water consumption, while a linear equation was established between them, in which the correlation coefficient and R^2 were above 0.6, which reached a more credible degree. Finally, we estimated the future ecological and

Table 7
Ecology and agricultural water consumption estimation (unit:10^8 m^3/year).

Type of water consumption	Baseline 2015	Linear-2035	Markov-2035
Ecological water	0.170	0.190	0.127
Agricultural water	0.343	0.458	0.394

agricultural water consumption based on the linear equation and the natural vegetation area and farmland area under different scenarios in 2035. The results showed that compared to the Baseline-2015 scenario, the ecological water use was 0.190 × 10^8 m^3/year in the Linear-2035 scenario, while the agricultural water use was 0.458 × 10^8 m^3/year in the Linear-2035 scenario. In the Linear-2035 scenario, the ecological water consumption increased by 0.020 × 10^8 m^3/year and the agricultural water consumption increased by 0.115 × 10^8 m^3/year. Compared to the Baseline-2015 scenario, the ecological water consumption in the Markov-2035 scenario was 0.127 × 10^8 m^3/year, while the agricultural water consumption in the Markov-2035 scenario was 0.394 × 10^8 m^3/year. The ecological water consumption decreased by 0.043 × 10^8 m^3/year and the agricultural water consumption increased by 0.051 × 10^8 m^3/year under the Markov-2035 scenario.

4. Discussion

4.1. Analysis the underlying driving of land cover change

The PLUS LEAS module was used in this work to drive the analysis of land cover changes from 2000 to 2006, 2006 to 2010, 2010 to 2015, and 2015 to 2020, and to generate spatially probabilistic maps of the development of each class (Fig. 9 & Fig. 10). We discovered from the spatial development probability map of each class that the spatial development probability of saline land was concentrated around the coccyx lake and continued to the east; the spatial development probability of agricultural land was concentrated in the plain area on both sides of the river; the spatial development probability of grassland was distributed spatially; and the spatial development probability of forestland was distributed spatially.

The probability of spatial development of saline land varied in degree in different periods, with the highest degree of development probability in 2015–2020, followed by 2000–2006, and finally 2010–2015. The spatial development probability of agricultural land increased in chronological order. The higher spatial development probability of grassland was concentrated in the southwest part of the area in 2000–2006, the high values in 2010–2015 were mainly concentrated in the north of the basin, and the high values in 2015–2020 were mainly concentrated in the east. The spatial development probability of forest varied in degree in different periods and basically grew in mountainous areas, with the highest development probability in 2010–2015, followed by 2015–2020, and finally in 2000–2006.

In each period, all the causes of land cover/use change were elevation dependent. Solar radiation, proximity to the river, slope, and precipitation were among the major factors on the fluctuation of water bodies. Temperature, proximity to the railroad, slope, pH, and solar radiation were the key factors of salinity change. Population, temperature, slope, pH, distance from roads, distance from government, precipitation, and solar radiation were the primary drivers of agricultural land change, with human activities accounting for a relatively substantial fraction of these elements. Solar radiation, precipitation, population, proximity from road, and soil texture were the primary driving forces behind changes in bare land. Precipitation, GDP, temperature, slope, proximity to roads, and population were the primary drivers of change in grassland and scrub. Population, total soil K and P, precipitation, soil organic matter content, and proximity to the river were the primary drivers of forest change. Population, slope, proximity from roads, GDP, proximity from government, and precipitation were the primary driving variables behind changes in glaciers and permanent

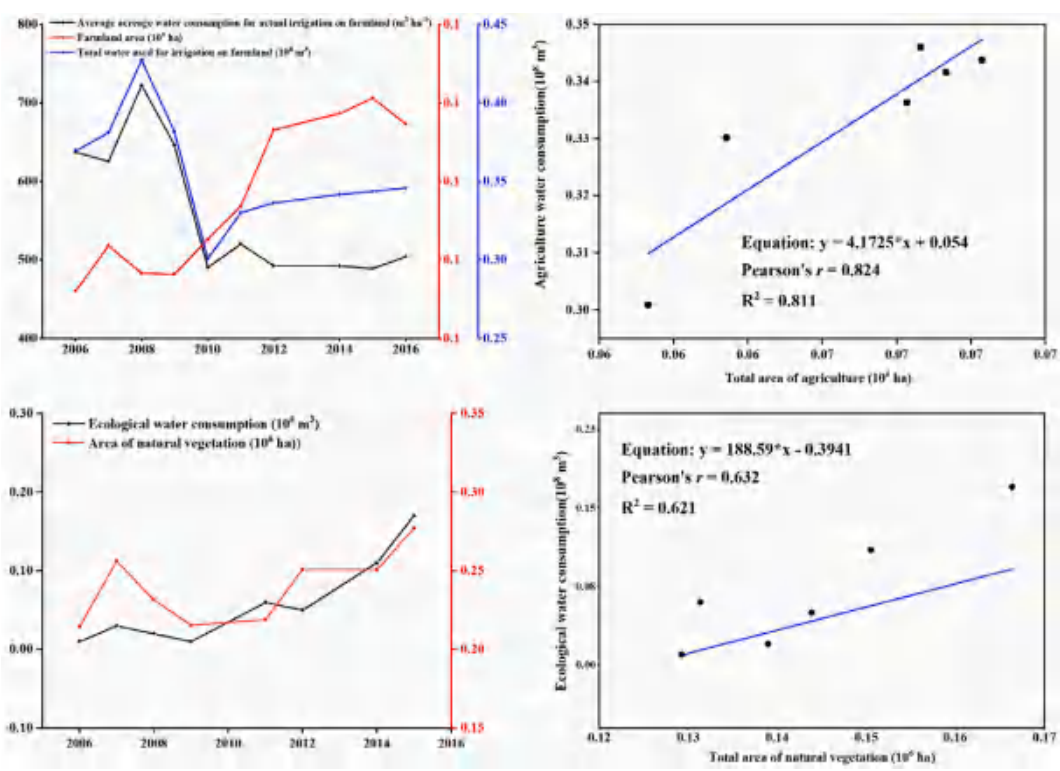

Fig. 8. Linear relationships and temporal trends in agricultural and ecological water consumption. (a) Time series trends in average acreage of agricultural water consumption, agricultural land area, and total agricultural water consumption. (b) Linear relationship between agricultural water consumption and agricultural land area. (c) Time series trend of ecological water consumption and natural vegetation area; (d) Linear relationship between ecological water consumption and natural vegetation area.

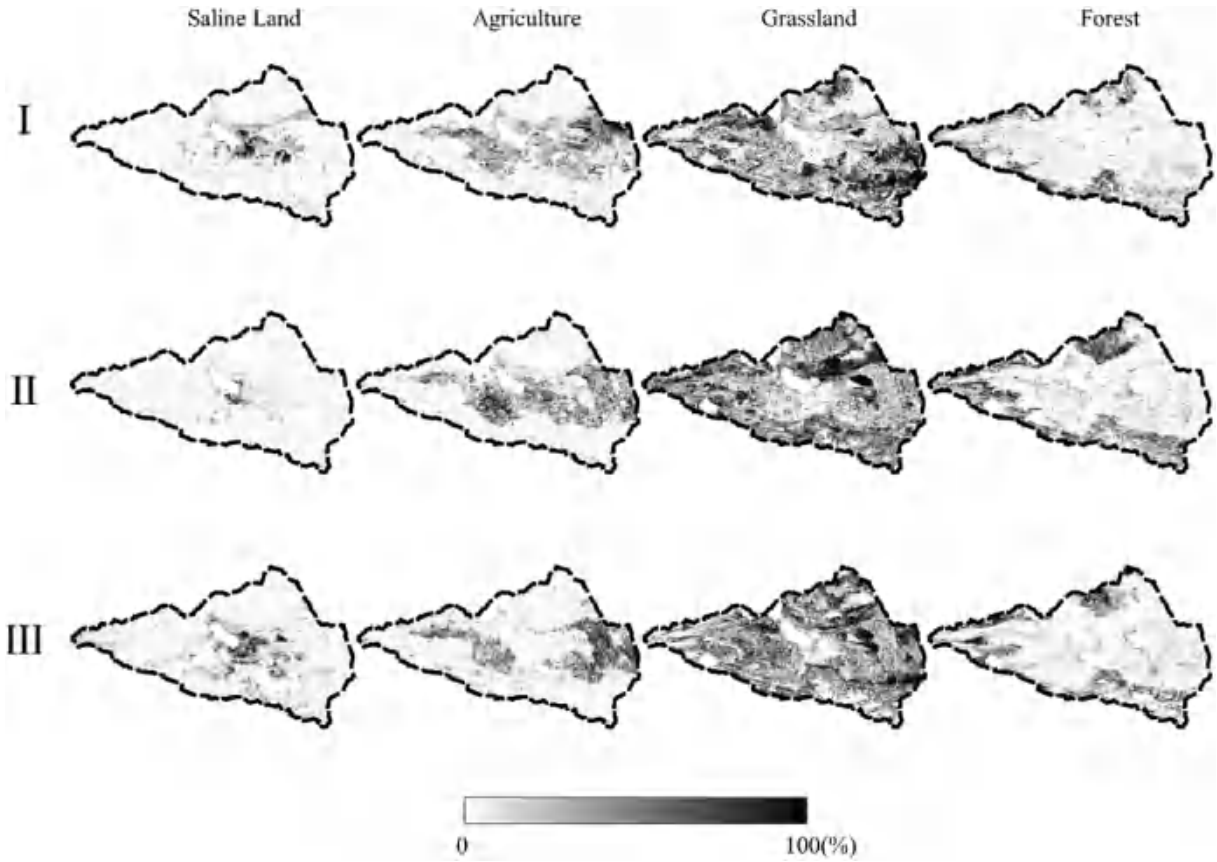

Fig. 9. Spatial expansion probabilities of major land types in different periods. I: for 2000 – 2006. II: for 2010 – 2015. III: for 2015 – 2020.

snowpack, with human activities having a stronger impact on glacier and snowpack ablation. As previously stated, the effect of increased human activity on land cover/use change cannot be overstated and has gradually become a key influence in changing local land cover types.

4.2. Analysis of agricultural water consumption changes

By reviewing the socio-economic statistical yearbook of Bortala Mongol Autonomous Prefecture, we found that the scale of local cotton cultivation had increased continuously in the past 17 years, from 413,100 ha in 2000 to 1,528,400 ha in 2017, an increase of 270%. The largest proportion of cotton cultivation area was followed by grain crops, wolfberry, and vegetables. Especially in 2010, the cotton planting area increased by 22.30% compared with the previous year, which was the largest increase in the study period, and at the same time in 2010, the large-scale use of high-efficiency water-saving drip irrigation technology began, in which the high-tech water-saving irrigation area increased by 59.30% compared with last year, which greatly reduced the average acreage of agricultural irrigation water. Cotton belongs to drought-tolerant plants, suitable for growth under more adequate light conditions, and relevant research shows that cotton requires far less water to grow than other food crops. Due to the rapid development of cotton cultivation and high-efficiency water-saving drip irrigation technology, we could observe in Fig. 9(a) that after 2010, although both agricultural water consumption and average per acre irrigation water consumption showed an increasing trend, the difference between the two magnitudes was obvious, and the increasing trend of agricultural water consumption was significantly higher than the average per acre irrigation water consumption, while the average per acre irrigation water consumption had a slight decreasing trend.

Based on the estimated agricultural water consumption under the different scenarios, we found that agricultural water consumption increased more in the Linear-2035 scenario than in the Markov-2035 scenario compared to the Baseline-2015 scenario. This was because the linear model used the average trend of all the study periods and ignored the decreasing trend of agricultural water consumption after 2010, thus resulting in a significant increase in agricultural water consumption in 2035. The increase in agricultural water consumption in the Markov-2035 scenario was slight compared to the Baseline-2015 scenario and much smaller than the Linear-2035 scenario, due to the Markov chain state Results were obtained by constructing a pre-state function to reflect small state changes in the variables in the model, so the results were reasonable.

Fig. 11 shows the time series of the various land cover types predicted by the Markov and linear models. We obtain a decreasing trend of 0.40 km2/year from the baseline, and the linear model predicts a decreasing trend of 0.40 km2/year in the area of the water bodies compared to the Markov model in 2035. The Markov prediction model is fundamentally different from the linear prediction model in that the former is a method of predicting future changes at each moment (or period) based on the state of the land cover type at the previous moment, while the latter is a method of predicting future changes based on the trend of the entire historical time series data. It can also be obtained from the figure that all linear prediction models obtain land cover areas with monotonically increasing or decreasing trends, while Markov prediction models obtain results without significant increasing or decreasing trends. We can also extend here, regarding the comparison of the results of linear and nonlinear prediction models. Since land cover change is not only influenced by a single factor in the reality world, but often by a combination of natural, human and policy activities, it is difficult for the traditional linear model to explain the dynamic land cover change, considering that Markov method is based on probability theory as a nonlinear means, so it can make reasonable prediction of future land cover change to a certain extent. Our results also show that

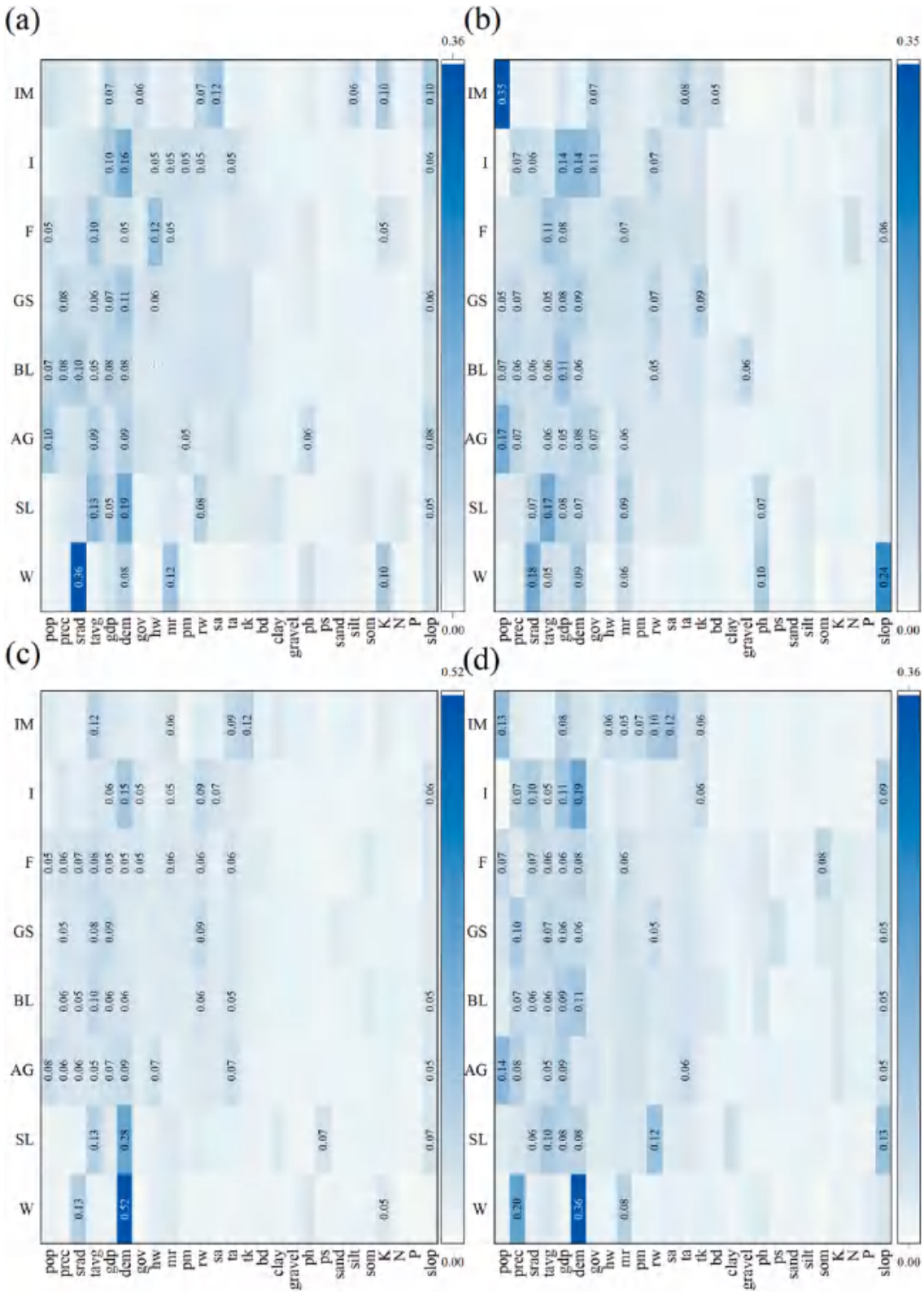

Fig. 10. Importance of each driving factor's contribution to the growth of the eight land cover types over four periods. Where W, SL, AG, BL, GS, F, I and IM indicates water bodies, saline land, agricultural land, bare land, grassland, scrub, forest, glacier and impervious layer, respectively. where pop, prec, srad, tavg, gdp, dem, gov, hw, mr, pm, rw, sa, ta, tk, bd, clay, gravel, ph, ps, sand, silt, som, K, N, P and Slop denotes average population density per kilometer, average annual precipitation, average annual solar radiation, average annual temperature, gross national product, elevation, proximity to government, proximity to highway, proximity to major rivers, proximity to primary road, distance from railroads, proximity to secondary roads, proximity to tertiary roads, proximity to trunk road, Soil Bulk Density, Soil Clay Content, Soil Gravel Content, Soil PH, Soil Porosity, Soil Sand Content, Soil Silt Content, Soil Organic Matter, Soil Total K, Soil Total N, Soil Total P and slope, respectively. (a) Shows 2000–2006; (b) shows 2006–2010; (c) shows 2010–2015; (d) shows 2015–2020.

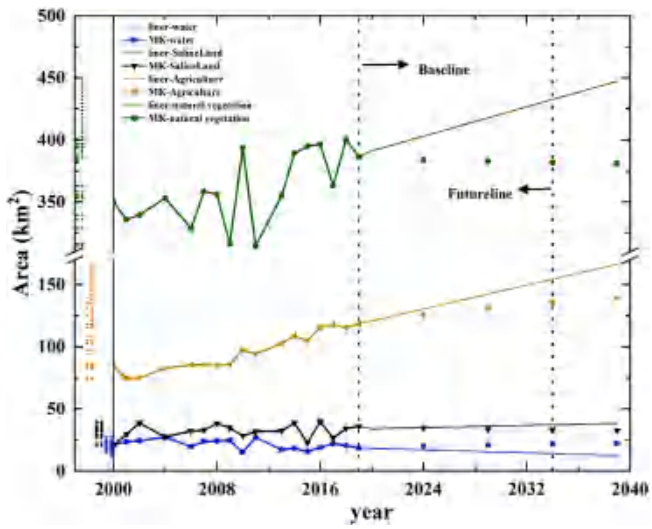

Fig. 11. Differences between linear and Markovian prediction methods.

the land cover area predicted by Markov coincides with the change pattern of historical land cover area, which is not a monotonically decreasing and increasing trend in the mathematical sense, and better reflects the state of future changes of real-world land cover. Some scholars have also explored the applicability of Markov methods for predicting land cover, and studies have found that Markov methods achieve more reliable estimates of land cover change, and a study by found that Markov-CA methods are advantageous in predicting future land use in arid areas (Halmy et al., 2015; Khwarahm et al., 2021; López et al., 2001; Zhang et al., 2021b). Therefore, the prediction of land cover change in this study may be more realistic by the Markov method.

5. Conclusion

Based on the land cover classification of the Lake Ebinur basin for the period 2000–2020, this study discovered that the area growth rates of saline, agricultural land, and natural vegetation (grassland, scrub, and forest) in the Lake Ebinur basin were 10.35 km^2/year, 111.24 km^2/year, and 161.09 km^2/year, respectively, while glacier and snow area decreased by 61.75 km^2/year.

The results of PLUS scenario simulation showed that the area of water bodies decreased under Linear-2035 scenario and increased under Markov-2035 scenario; the area of saline land, agricultural land and natural vegetation showed an increase under both scenarios; glaciers and permanent snow and bare land showed a decreasing trend under both scenarios. The results of ecological water use and agricultural water consumption under future scenario estimation showed that ecological water consumption showed an increasing trend under Linear-2035 scenario and a decreasing trend under Markov-2035 scenario. Agricultural water consumption showed a significant increasing trend in both scenarios, 0.458×10^8 m^3 in the Linear-2035 scenario and 0.394×10^8 m^3 in the Markov-2035 scenario. The results of land cover driving factor analysis showed that the main driving factors of saline land change were: dem, tavg, slop, mr, rw and ph; the main driving factors of agricultural land change were pop, gdp, dem and prec; the main driving factors of natural vegetation change were: dem, tk, rw, som, tavg and hw.

According to the findings of this study, future regional policymakers must gradually raise water allocation for agriculture based on a minimum ecological water supply to accomplish sustainable growth in arid region basins.

CRediT authorship contribution statement

Qingling Bao: Software, Visualization, Writing – original draft, Writing – review & editing. **Jianli Ding:** Funding acquisition, Supervision. **Lijing Han:** Methodology. **Jiang Li:** Writing – review & editing. **Xiangyu Ge:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The National Natural Science Foundation of China (Grant No. 42171269), and the project of Academician Workstation in Xinjiang Uygur Autonomous Region, China (Grant No. 2020.B-001) supported this work.

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